

1. Administrative & Study Identification

Full Study Title: A Phase 1 Study of PIT565, a First-in-Class, Anti-CD3, Anti-CD19, Anti-CD2 Trispecific Antibody in Patients with Relapsed and/or Refractory B-Cell Malignancies **Short Title/ Acronym:** Study of PIT565 in Relapsed and/or Refractory B-cell Malignancies **Protocol Number:** CPIT565A12101 **NCT Number:** NCT05397496 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** PIT565 (CD3 × CD2 × CD19 Trispecific Antibody) **Phase of Development:** Phase I **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety and tolerability of PIT565 and to identify maximal tolerated doses (MTDs) and/or recommended doses (RDs), schedule, and route of administration in relapsed and/or refractory B-cell Non-Hodgkin lymphoma (R/R B-NHL) and relapsed and/or refractory B-cell acute lymphoblastic leukemia (R/R B-ALL). **Secondary Objectives:** To evaluate preliminary anti-tumor activity (including overall response rate [ORR], complete response [CR] rate, best overall response [BOR], duration of response [DOR], progression-free survival [PFS] for B-NHL, event-free survival [EFS] for B-ALL, and overall survival [OS]), and to assess the pharmacokinetics (PK) and immunogenicity of PIT565.

2.2 Background & Rationale There is a significant unmet therapeutic need for patients with R/R B-NHL and R/R B-ALL who have relapsed or failed multiple lines of therapy. PIT565 is an investigational, first-in-class, trispecific antibody that targets CD19 on malignant B cells while simultaneously engaging CD3 and CD2 (a costimulatory receptor) on T cells. This dual engagement redirects T-cell cytotoxicity toward CD19-positive malignant B cells. Activation of CD2 signaling prevents the transcriptional changes characteristic of T-cell exhaustion, thus promoting a more potent and sustained anti-tumor immune response compared to traditional CD3 bispecific antibodies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Adaptive dose-escalation and dose-expansion)

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 140 patients **Number of Sites:** Multicenter, global (including sites in Belgium, France, Israel, Italy, Japan, Singapore, Spain, and the USA) **Estimated Study Start:** 26-May-2022 **Estimated Primary Completion:** 30-Jan-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 . 2. Documented diagnosis of R/R B-NHL or R/R CD19-positive B-ALL. 3. Must have relapsed after or failed to respond to at least two prior lines of therapy (including an anti-CD20 monoclonal antibody-containing chemotherapy combination regimen for patients with R/R B-NHL). 4. Must have at least one bi-dimensionally measurable nodal/extranodal lesion on PET/CT (for B-NHL) and/or morphologic disease in the bone marrow ($\geq 5\%$ blasts) (for B-ALL).

4.2 Exclusion Criteria 1. Active central nervous system (CNS) involvement by malignancy, symptomatic CNS metastases requiring local therapy, or increasing doses of corticosteroids within 2 weeks prior to study treatment. 2. Active, known, or suspected autoimmune disease (exceptions include vitiligo, residual hypothyroidism requiring hormone replacement, or psoriasis not requiring systemic treatment). 3. Patients receiving systemic treatment with any immunosuppressive medication.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered either once weekly (Q1W) or once every 2 weeks (Q2W), with and without a priming dose, over a 28-day cycle. Dose escalation is guided by an adaptive Bayesian Logistic Regression Model (BLRM). **Route of Administration:** Intravenous (IV) or Subcutaneous (SC). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. **Prohibited:** Systemic immunosuppressive medications and escalating doses of corticosteroids during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline PET/CT or Bone Marrow Biopsy, Safety Labs
Baseline	Day 1 (Cycle 1)	First Dose of PIT565, PK/PD Sampling, Vital Signs	Interim Weekly/Bi-weekly Safety

Assessments, TEAE Monitoring, Efficacy Assessment (Lugano Criteria for NHL / NCCN Guidelines for ALL) | | **End of Study** | At Progression | Final Tumor Assessment, Exit Interview, Safety Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, frequency, and severity of treatment-emergent adverse events (TEAEs) and dose-limiting toxicities (DLTs) to determine the Maximum Tolerated Dose (MTD) and Recommended Dose (RD). **Measurement Method:** Clinical evaluation, standard laboratory assessments, and vital sign monitoring per CTCAE criteria.

7.2 Secondary & Exploratory Endpoints 1. Overall Response Rate (ORR), Complete Response (CR) rate, and Best Overall Response (BOR). 2. Progression-Free Survival (PFS), Event-Free Survival (EFS), Overall Survival (OS), and Duration of Response (DOR). 3. Pharmacokinetics (PK) and immunogenicity profiles of PIT565.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of TEAEs, with specific protocolized assessments for Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS) common with T-cell redirecting therapies. **Serious Adverse Events (SAEs):** Typically reported within 24 hours to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** An internal safety review committee guides the dose-escalation phase utilizing the Escalation with Overdose Control (EWOC) principle.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of study medication) and Efficacy Evaluable Set. **Sample Size Justification:** Approximately 140 patients. Sample size in the dose-escalation phase is adaptive and determined by the BLRM to ensure safety while identifying the MTD/RD. **Handling of Missing Data:** Efficacy data is analyzed as observed; missing data is generally not imputed except for standard conventions applied to partial dates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to ensure

protocol compliance and data integrity. **Audit Trail:** Verified through electronic Case Report Form (eCRF) locking and robust data clarification workflows.

11. Study Identifiers & Governance

Primary ID: CPIT565A12101 **Registry Number:** NCT05397496 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Study of PIT565 in Relapsed and/or Refractory B-cell Malignancies **Official Regulatory Title:** An open-label, multicenter, phase I study of PIT565 in patients with R/R B-NHL and R/R B-ALL

12. Clinical Framework (Oncology Specific)

Primary Condition: Relapsed and/or Refractory B-cell Malignancies (B-NHL and B-ALL) **Diagnosis Threshold:** Morphologic/radiographic evidence of disease relapse or progression following ≥ 2 prior lines of therapy. **Medical Methodology:** T-cell Redirecting Immunotherapy (Trispecific Antibody) **Optimization Target:** CD3 x CD2 x CD19 engagement for targeted T-cell mediated cytotoxicity.

13. Study Procedures & Methodology

Management Pathway: Stepwise dose-escalation followed by indication-specific dose expansion cohorts. **Education Protocol (HCP):** Site initiation and training focused on the administration of PIT565 and the specialized management of CRS and ICANS. **Education Protocol (Patient):** Informed consent detailing expected toxicities and the requirement for rapid reporting of neurological or systemic symptoms. **Quality Audit Mechanism:** Continuous safety monitoring guided by the BLRM criteria and regular sponsor audits.

14. Observation & Follow-Up Schedule

Total Duration: Ongoing until disease progression, intolerable toxicity, or patient withdrawal. **Onsite Visit Cadence:** Highly frequent during Cycle 1 (Q1W or Q2W dosing) with tapering frequency in subsequent cycles. **Remote Monitoring Frequency:** Inter-cycle assessments tailored to patient safety and logistical requirements. **Checklist Review Interval:** Prior to the administration of every dose to ensure toxicity resolution.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |